1. If the coefficients of x^7 & x^8 in the expansion of $\left[2+\frac{x}{3}\right]^n$ are equal, then the value of n is :

यदि $\left[2+\frac{x}{3}\right]^n$ के प्रसार में यदि x^7 व x^8 का गुणांक बराबर हो, तो n का मान होगा :

(A) 15 (B) 45 (C) 55 (D) 56

Ans. C

Sol. $2^n\left(1+\frac{x}{6}\right)^n \Rightarrow T_{r+1}=2^n\,.\,{}^nC_r\left(\frac{x}{6}\right)^r \Rightarrow 2^n\,.\,{}^nC_7\,.\,\frac{1}{6^7}=2^n\,.\,{}^nC_8\,.\,\frac{1}{6^8}$

$\Rightarrow 6\,.\,{}^nC_7={}^nC_8 \Rightarrow n-7=48 \Rightarrow n=55$]

2. The coefficient of x^{49} in the expansion of $(x-1)\left(x-\frac{1}{2}\right)\left(x-\frac{1}{2^2}\right).....\left(x-\frac{1}{2^{49}}\right)$ is equal to :

(A) $-2\left(1-\frac{1}{2^{50}}\right)$ (B) coefficient of x

(C) – (coefficient of x) (D) $-2\left(1-\frac{1}{2^{49}}\right)$

$(x-1)\left(x-\frac{1}{2}\right)\left(x-\frac{1}{2^2}\right).....\left(x-\frac{1}{2^{49}}\right)$ के प्रसार में x^{49} का गुणांक होगा

(A) $-2\left(1-\frac{1}{2^{50}}\right)$ (B) x का गुणांक

(C) –(x का गुणांक) (D) $-2\left(1-\frac{1}{2^{49}}\right)$

Ans. A

Sol. coeff.. of x^{49} in this series is

$$-\left[1+\frac{1}{2}+\frac{1}{2^2}+.....+\frac{1}{2^{49}}\right] = -\left[\frac{1-\frac{1}{2^{50}}}{1-\frac{1}{2}}\right] = -2.\left[1-\frac{1}{2^{50}}\right]$$

3. The sum of the binomial coefficients of $\left[2x+\frac{1}{x}\right]^n$ is equal to 256. The constant term in the expansion is :

यदि $\left[2x+\frac{1}{x}\right]^n$ के प्रसार में द्विपद गुणांकों का योग 256 है, तो इसमें अचर पद का मान होगा :

(A) 1120 (B) 2110 (C) 1210 (D) none

Ans. A

Sol. $C_0 + C_1 + C_2 + \ldots\ldots C_n = 256 \Rightarrow 2^n = 2^8 \Rightarrow n = 8$

4. In the expansion of $\left(3^{-\frac{x}{4}} + 3^{\frac{5x}{4}}\right)^n$ the sum of the binomial combinatorial coefficients is 64 and the term with the greatest binomial coefficient exceeds the third term by (n – 1), then the value of x must be :

$\left(3^{-\frac{x}{4}} + 3^{\frac{5x}{4}}\right)^n$ के प्रसार में द्विपद गुणांकों का योग 64 है तथा महत्तम द्विपद गुणांकों वाले पद तथा तीसरे पद का अन्तर (n – 1) है x का मान होगा:

(A) 1 (B) 2 (C) 0 (D) – 1

Ans. C

Sol. $C_0 + C_1 + \ldots\ldots + C_n \quad \Rightarrow \quad 2^n = 2^6 \quad \Rightarrow \quad n = 6$

5. In the expansion of $\left(\frac{x+1}{x^{2/3} - x^{1/3} + 1} - \frac{x-1}{x - x^{1/2}}\right)^{10}$, the term which does not contain x is :

(A) ${}^{10}C_0$ (B) ${}^{10}C_7$ (C) ${}^{10}C_4$ (D) none

श्रेणी $\left(\frac{x+1}{x^{2/3} - x^{1/3} + 1} - \frac{x-1}{x - x^{1/2}}\right)^{10}$ के प्रसार में x से स्वतंत्र पद होगा :

(A) ${}^{10}C_0$ (B) ${}^{10}C_7$ (C) ${}^{10}C_4$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\frac{\left(x^{1/3}\right)^3 + 1}{x^{2/3} - x^{1/3} + 1} = x^{1/3} + 1;$ and $\frac{x-1}{\sqrt{x}\left(\sqrt{x} - 1\right)} = \frac{\sqrt{x}+1}{\sqrt{x}} = 1 + x^{-1/2}$

hence binomial expansion is $(x^{1/3} - x^{-1/2})^{10}$ now proceed

6. Co-efficient of α^t in the expansion of,
$(\alpha + p)^{m-1} + (\alpha + p)^{m-2}(\alpha + q) + (\alpha + p)^{m-3}(\alpha + q)^2 + \ldots\ldots (\alpha + q)^{m-1}$ where $\alpha \neq -q$ and $p \neq q$ is :
श्रेणी $(\alpha + p)^{m-1} + (\alpha + p)^{m-2}(\alpha + q) + (\alpha + p)^{m-3}(\alpha + q)^2 + \ldots\ldots (\alpha + q)^{m-1}$ में α^t का गुणांक होगा जहाँ $\alpha \neq -q$ तथा $p \neq q$:

(A) $\frac{{}^mC_t\left(p^t - q^t\right)}{p-q}$ (B) $\frac{{}^mC_t\left(p^{m-t} - q^{m-t}\right)}{p-q}$

(C) $\frac{{}^mC_t\left(p^t + q^t\right)}{p-q}$ (D) $\frac{{}^mC_t\left(p^{m-t} + q^{m-t}\right)}{p-q}$

Ans. B

Sol. $E = (\alpha + p)^{m-1}\left[1 + \frac{\alpha+q}{\alpha+p} + \left(\frac{\alpha+q}{\alpha+p}\right)^2 + \ldots\ldots + \left(\frac{\alpha+q}{\alpha+p}\right)^{m-1}\right] = (\alpha + p)^{m-1}\left[\frac{\left(\frac{\alpha+q}{\alpha+p}\right)^m - 1}{\left(\frac{\alpha+q}{\alpha+p}\right) - 1}\right]$

$= \frac{(\alpha+p)^m[(\alpha+p)^m - (\alpha+q)^m]}{(\alpha+p)^m(p-q)}$

hence coefficient of α^t in $\frac{(\alpha+p)^m-(\alpha+q)^m}{p-q} = \frac{(p+\alpha)^m-(q+\alpha)^m}{p-q} = \frac{{}^mC_t\left(p^{m-t}-q^{m-t}\right)}{p-q}$

7. If the second term of the expansion $\left[a^{1/13}+\frac{a}{\sqrt{a^{-1}}}\right]^n$ is $14a^{5/2}$ then the value of $\frac{{}^nC_3}{{}^nC_2}$ is :

$\left[a^{1/13}+\frac{a}{\sqrt{a^{-1}}}\right]^n$ के प्रसार में दूसरा पद $14a^{5/2}$ हो, तो $\frac{{}^nC_3}{{}^nC_2}$ का मान होगा :

(A) 4 (B) 3 (C) 12 (D) 6

Ans. A

Sol. $T_2 = {}^nC_1\left(a^{1/13}\right)^{n-1}\cdot a\sqrt{a} = 14a^{5/2}$ or $n\cdot a^{\frac{n-1}{13}} = 14a$ or $n\cdot a^{\frac{n-14}{13}} = 14$

hence $\frac{n-14}{13} = 0 \Rightarrow n = 14.$ Now $\frac{{}^{14}C_3}{{}^{14}C_2} = \frac{14!}{3!\cdot 11!}\frac{2!\cdot 2!}{14!} = \frac{12}{3} = 4$

8. The positive value of 'a' so that the coefficient of x^5 is equal to that of x^{15} in the expansion of $\left(x^2+\frac{a}{x^3}\right)^{10}$ is :

$\left(x^2+\frac{a}{x^3}\right)^{10}$ के विस्तार में 'a' का मान क्या होगा जबकि x^5 और x^{15} का गुणांक समान हो :

(A) $\frac{1}{2\sqrt{3}}$ (B) $\frac{1}{\sqrt{3}}$ (C) 1 (D) $2\sqrt{3}$

Ans. A

Sol. $\left(x^2+\frac{a}{x^3}\right)^{10} = \sum_{r=0}^{10}{}^{10}C_r\left(x^2\right)^{10-r}\cdot\frac{a^r}{x^{3r}}$

$= \sum_{r=0}^{10}{}^{10}C_r x^{20-5r}\cdot a^r$

Coefficient of x^5 = coefficient of x^{15}

$\Rightarrow {}^{10}C_3\, a^3 = {}^{10}C_1\, a \Rightarrow 120a^3 = 10a \Rightarrow 12a^3 - a = 0$

$\Rightarrow a = 0$ or $a = \pm\frac{1}{\sqrt{12}}$

Given $a > 0$

$\Rightarrow a = \frac{1}{\sqrt{12}} = \frac{1}{2\sqrt{3}}$

9. Number of rational terms in the expansion of $\left(\sqrt{2}+\sqrt[4]{3}\right)^{100}$ is :

$\left(\sqrt{2}+\sqrt[4]{3}\right)^{100}$ के प्रसार में कुल परिमेय पदों की संख्या होगी :

(A) 25 (B) 26 (C) 27 (D) 28

Ans. B

Sol. $T_{r+1} = {}^{100}C_r \; 2^{\frac{100-r}{2}} \cdot 3^{r/4}$

$\Rightarrow$ r must be even and divisible by 4 $\Rightarrow$ r = 0, 4, 8, ... , 100

10. In the expansion of $\left(\sqrt{\frac{q}{p}}+\sqrt[10]{\frac{p^7}{q^3}}\right)^n$, there is a term similar to pq, then that term is equal to :

$\left(\sqrt{\frac{q}{p}}+\sqrt[10]{\frac{p^7}{q^3}}\right)^n$ के प्रसार में pq जैसे पद का मान होगा :

(A) 210 pq (B) 252 pq (C) 120 pq (D) 45 pq

Ans. B

Sol. $5n - 8r = 10$ & $12r - n = 10 \Rightarrow 10 + 8r = 12r - 10$

$\Rightarrow$ $r = 5 \Rightarrow n = 10 \Rightarrow$ coeff. of pq $= {}^{10}C_5 = 252$

11. The term independent of 'x' in the expansion of $\left(9x - \frac{1}{3\sqrt{x}}\right)^{18}$, $x > 0$, is α times the corresponding binomial coefficient. Then 'α' is :

$\left(9x - \frac{1}{3\sqrt{x}}\right)^{18}$ के प्रसार में 'x' से मुक्त पद तथा उसी पद के द्विपद गुणांक का अनुपात यदि 'α' हो, तो α =

(यहाँ $x > 0$ है) :

(A) 3 (B) 1/3 (C) – 1/3 (D) 1

Ans. D

Sol. $T_{r+1} = {}^{18}C_r \cdot (9x)^{18-r} \cdot (-1)^r \cdot \frac{(x)^{-r/2}}{3^r}$

$T_{r+1} = {}^{18}C_r \cdot 9^{18-r} \cdot \frac{(-1)^r}{3^r} \cdot x^{\left(18-\frac{3r}{2}\right)}$

$\therefore \quad 18 - \frac{3r}{2} = 0 \quad \Rightarrow \quad r = 12$

$${}^{18}C_{12} \cdot \frac{9^6}{3^{12}} = \alpha \cdot {}^{18}C_{12} \Rightarrow \alpha = 1$$

12. Greatest term in the binomial expansion of $(a + 2x)^9$ when $a = 1$ & $x = 1/3$ is :

जब $a = 1, x = \frac{1}{3}$ हो, तो $(a + 2x)^9$ का महत्तम पद होगा :

(A) 3rd & 4th (B) 4th & 5th (C) only 4th (D) only 5th

Ans. B

Sol. $\frac{T_{r+1}}{T_r} = \frac{{}^9C_r a^{9-r}.(2x)^r}{{}^9C_{r-1} a^{9-r+1}(2x)^{r-1}} = \frac{9-r+1}{r}\ \frac{2x}{a} = \frac{9-r+1}{r}\ \frac{2}{3} = \frac{10-r}{r}\ \frac{2}{3}$

$\therefore\ T_{r+1} > T_r$ if $\frac{10-r}{r}\ \frac{2}{3} > 1$ i.e. $20 > 5r$ i.e. $r < 4$

Hence for values of 'r' upto 3, $T_{r+1} < T_r$

when $r = 4$, $T_{r+1} = T_r$ i.e. $T_4 = T_5$ & both of them are the greatest terms

13. The expression $[x + (x^3-1)^{1/2}]^5 + [x - (x^3-1)^{1/2}]^5$ is a polynomial of degree :

व्यंजक $[x + (x^3-1)^{1/2}]^5 + [x - (x^3-1)^{1/2}]^5$ कितने घात का बहुपद होगा :

(A) 5 (B) 6 (C) 7 (D) 8

Ans. C

Sol. Let $(x^3 - 1)^{1/2} = y$

$\therefore\ E = (x + y)^5 + (x - y)^5 = 2\,[{}^5C_0\, x^5 + {}^5C_2\, x^3y^2 + {}^5C_4\, xy^4]$

$E = 2\,[x^5 + 10\,x^3(x^3 - 1) + 5x(x^3 - 1)^2$

hence degree is 7

14. Sum of all the rational terms in the expansion of $\left(3^{\frac{1}{4}} + 4^{\frac{1}{3}}\right)^{12}$, is :

(A) 27 (B) 256 (C) 283 (D) none

$\left(3^{\frac{1}{4}} + 4^{\frac{1}{3}}\right)^{12}$ के प्रसार में सारे परिमेय पदों का योग होगा :

(A) 27 (B) 256 (C) 283 (D) इनमें से कोई नहीं

Ans. C

Sol. $T_{r+1} = {}^{12}C_r \left(3^{\frac{1}{4}}\right)^{12-r} \cdot (4)^{r/3}$

$= {}^{12}C_r \cdot 3^{(3-r/4)} \cdot (4)^{r/3}$

For rational term $r = 0$ or 12

$\therefore$ sum $= T_1 + T_{13} = {}^{12}C_0 \cdot 3^3 + {}^{12}C_{12} \cdot 4^4$

$= 3^3 + 4^4 = 27 + 256 = 283$

15. If the $(r+1)^{th}$ term in the expansion of $\left(\sqrt[3]{\frac{a}{\sqrt{b}}} + \sqrt{\frac{b}{\sqrt[3]{a}}}\right)^{21}$ contains a and b to one and the same power, then the value of r is :

यदि $\left(\sqrt[3]{\frac{a}{\sqrt{b}}}+\sqrt{\frac{b}{\sqrt[3]{a}}}\right)^{21}$ के प्रसार में $(r+1)$ वें पद में a तथा b की घात समान हो तो r का मान है

(A) 9 (B) 10 (C) 8 (D) 6

Ans. A

Sol. $\left(3\sqrt{\frac{a}{\sqrt{b}}}+\sqrt{\frac{b}{\sqrt[3]{a}}}\right)^{21}=\left[\left(\frac{a}{\sqrt{b}}\right)^{\frac{1}{3}}+\left(\frac{b}{a^{\frac{1}{3}}}\right)^{\frac{1}{2}}\right]^{21}$

$$=\sum_{r=0}^{21}{}^{n}C_{r}\left(\frac{a}{\sqrt{b}}\right)^{\frac{n-r}{3}}\cdot\left(\frac{b}{a^{\frac{1}{3}}}\right)^{\frac{r}{2}};n=21$$

$$=\sum_{r=0}^{21}{}^{n}C_{r}\cdot a^{\frac{n-r}{3}}\cdot b^{\frac{r-n}{6}}\cdot b^{\frac{r}{2}}\cdot a^{\frac{-r}{6}};n=21$$

$$=\sum_{r=0}^{21}{}^{n}C_{r}a^{\frac{2n-2r-r}{6}}\cdot b^{\frac{r-n+3r}{6}};n=21$$

$$=\sum_{r=0}^{21}{}^{n}C_{r}a^{\frac{2n-3r}{6}}b^{\frac{4r-n}{6}}$$

$$\frac{2n-3r}{6}=\frac{4r-n}{6}$$

$\Rightarrow 2n-3r=4r-n$

$\Rightarrow 3n=7r$

$\Rightarrow 7r=21\times 3$

$\Rightarrow r=9$

16. In the expansion of $(5^{1/2}+7^{1/8})^{1024}$, the number of integral terms is:

$(5^{1/2}+7^{1/8})^{1024}$ के प्रसार में पूर्णांक पदों की संख्या है:

(A) 128 (B) 129 (C) 130 (D) 131

Ans. B

Sol. $\left(5^{\frac{1}{2}}+7^{\frac{1}{8}}\right)^{1024}=\sum_{r=0}^{1024}{}^{1024}C_{r}\left(5^{\frac{1}{2}}\right)^{1024-r}(7)^{\frac{r}{8}}$

$$=\sum_{r=0}^{1024}{}^{1024}C_{r}(5)^{\frac{1024-r}{2}}7^{\frac{r}{8}}$$

For integral terms

$\frac{r}{8} \in I \ \& \ \frac{1024-r}{2} \in I$

$\Rightarrow r = 8k; \ 0 \le k \le 128$

$\Rightarrow$ Total integral terms is 129.

17. Find the number of different terms in the sum $(1+x)^{2009} + (1+x^2)^{2008} + (1+x^3)^{2007}$.
श्रेणी $(1+x)^{2009} + (1+x^2)^{2008} + (1+x^3)^{2007}$ में कुल कितने भिन्न पद होगें।

Ans. 4017

Sol. **Alternatively:** $n(A \cup B \cup C) = n(A) + n(B) + n(C) - \left(n(A \cap B) + n(B \cap C) + n(C \cap A) + n(A \cap B \cap C)\right)$

$= (2010 + 2009 + 2008) - \underbrace{1005}_{A \cap B} + \underbrace{670}_{B \cap C} + \underbrace{670}_{C \cap A} + 335 = 4017$

1. Given $(1-2x+5x^2-10x^3)(1+x)^n = 1+a_1x+a_2x^2+....$ and that $a_1^2 = 2a_2$ then the value of n is :

दया है $(1-2x+5x^2-10x^3)(1+x)^n = 1+a_1x+a_2x^2+....$ यदि $a_1^2 = 2a_2$ तो n का मान होगा ।:

(A) 6 (B) 2 (C) 5 (D) 3

Ans. A

Sol. $(1-2x+5x^2+10x^3)\ [C_0+C_1x+C_2x^2+....] = 1+a_1x+a_2x^2+.....$

$a_1 = n-2$ and $a_2 = \frac{n(n-1)}{2} - 2n + 5$

put $a_1^2 = 2a_2$

$(n-2)^2 = n(n-1) - 4n + 10$

$n^2 - 4n + 4 = n^2 - 5n + 10$

$n = 6$ Ans]

2. The coefficient of 1/x in the expansion of $(1+x)^n\left(1+\frac{1}{x}\right)^n$ is -

(A) $\frac{n!}{(n-1)!\,(n+1)!}$ (B) $\frac{(2n)!}{(n-1)!\,(n+1)!}$

(C) $\frac{(2n)!}{(2n-1)!\,(2n+1)!}$ (D) None of these

$(1+x)^n\left(1+\frac{1}{x}\right)^n$ के प्रसार में 1/x का गुणांक है

(A) $\frac{n!}{(n-1)!\,(n+1)!}$ (B) $\frac{(2n)!}{(n-1)!\,(n+1)!}$

(C) $\frac{(2n)!}{(2n-1)!\,(2n+1)!}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $(1+x)^n\cdot\left(1+\frac{1}{x}\right)^n = \frac{(1+x)^{2n}}{x^n}$

$\because (1+x)^{2n} = {}^{2n}C_0 + {}^{2n}C_1x + ... + {}^{2n}C_nx^n + {}^{2n}C_{n+1}x^{n+1} + ... + {}^{2n}C_{2n}x^{2n}$

Coefficient of $\frac{1}{x}$ in the expansion of $(1+x)^n\cdot\left(1+\frac{1}{x}\right)^n$ is ${}^{2n}C_{n-1}$

$= \dfrac{\lfloor 2n}{\lfloor n-1 \lfloor n+1}$

3. If $6^{83} + 8^{83}$ is divided by 49, then the remainder is :
यदि $6^{83} + 8^{83}$ को 49 से भाग दिया जाये तो शेषफल होगा:
(A) 35 (B) 5 (C) 1 (D) 0

Ans. A

Sol. $(1 + 7)^{83} + (7 - 1)^{83} = (1 + 7)^{83} - (1 - 7)^{83}$
$= 2[{}^{83}C_1 \cdot 7 + {}^{83}C_3 \cdot 7^3 + ...+ {}^{83}C_{83} \cdot 7^{83}] = (2 \cdot 7 \cdot 83) + 49I$ where I an integer
$\therefore \quad 14 \times 83 = 1162$

$$\frac{1162}{49} = 23\frac{35}{49}$$

$\therefore$ remainder is 35

4. The remainder, when $(15^{23} + 23^{23})$ is divided by 19, is :
जब $(15^{23} + 23^{23})$ को 19 से भाग दिया जाये तो शेषफल होगा:
(A) 4 (B) 15 (C) 0 (D) 18

Ans. C

Sol. $E = (19 - 4)^{23} + (19 + 4)^{23}$
$= 2\,[19^{23} + {}^{23}C_2 \cdot 19^{21} \cdot 4^2 + ... + {}^{23}C_{22} \cdot 19 \cdot 4^{22}]$
$= 2 \cdot 19\,[19^{22} + {}^{23}C_2 \cdot 19^{20} \cdot 4^2 + ...+ {}^{23}C_{22} \cdot 4^{22}\,]$
$\Rightarrow$ E is divisible by 19 $\Rightarrow$ Remainder = 0

5. Last three digits of the number $N = 7^{100} - 3^{100}$ are :
$N = 7^{100} - 3^{100}$ के अन्तिम तीन अंक होगें :
(A) 100 (B) 300 (C) 500 (D) 000

Ans. D

Sol. onsider $(5 + 2)^{100} - (5 - 2)^{100}$
$= 2\,[{}^{100}C_1\, 5^{99} \cdot 2 + {}^{100}C_3\, 5^{97} \cdot 2^3 + ... + {}^{100}C_{99} 5 \cdot 2^{99}]$
$= 2\,[1000 \cdot 5^{98} + {}^{100}C_3\, 5^{94} + ... + 1000 \cdot 2^{98}]$
$\Rightarrow$ minimum 000 as last three digits. $\Rightarrow$ (D)

6. **Statement-1:** $11^{25} + 12^{25}$ when divided by 23 leaves the remainder zero.
because
Statement-2: $a^n + b^n$ is divisible by $(a + b)\ \forall\ n \in N$.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.
कथन-1 : $11^{25} + 12^{25}$ को जब 23 से भाग दिया जाता है, तो यह शेषफल शून्य देता है।
क्योंकि
कथन-2 : $a^n + b^n$, $(a + b)$ से विभाजित होगा $n \in N$.
(A) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण है।
(B) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण **नहीं** है।
(C) कथन–1 सत्य है, कथन–2 असत्य है।

(D) कथन–1 असत्य है, कथन– 2 सत्य है।

Ans. C

Sol. $a^n + b^n$ is divisible by $(a + b)$ only if n is odd

7. When 2^{301} is divided by 5, the least positive remainder is :
2^{301} को 5 से विभाजित करने पर निम्नतम् धनात्मक शेषफल है :
(A) 4 (B) 8 (C) 2 (D) 6

Ans. C

Sol. $2^{301} = 2 \cdot 2^{300} = 2 \cdot (4)^{150}$

$= 2 \cdot (5 - 1)^{150}$

$$= 2\sum_{r=0}^{150} {}^{150}C_r (5)^{150-r} \cdot (-1)^r$$

$$= 2\left[{}^{150}C_0 5^{150} (-1)^0 + ... + {}^{150}C_{149} 5^1 \cdot (-1)^{149} \right] + 2\left[{}^{150}C_{150} (-1)^{150} 5^0 \right]$$

$= 5\lambda + 2; \lambda \in I^+$

$\Rightarrow$ Remiander = 2

8. Last two digits of 21 to the $(100)^{th}$ power, is :
$(21)^{100}$ के अन्तिम दो अंक होगें :
(A) 00 (B) 01 (C) 11 (D) 21

Ans. B

Sol. $(21)^{100} = (1 + 20)^{100} = 1 + {}^{100}C_1 \cdot 20 + {}^{100}C_2 \cdot 20^2 + ... + {}^{100}C_{100} \cdot 20^{100}$
hence last two digit are 01

9. The remainder, if $1 + 2 + 2^2 + 2^3 + ...+2^{1999}$ is divided by 5 is :
$1 + 2 + 2^2 + 2^3 + ...+2^{1999}$ को 5 से भाग देने पर शेषफल होगा:
(A) 0 (B) 1 (C) 2 (D) 3

Ans. A

Sol. $\frac{1(2^{2000} - 1)}{1} = 2^{2000} - 1$

$(5 - 1)^{1000} - 1 = (1 - 5)^{1000} - 1$
$1 - {}^{1000}C_1 \cdot 5 + {}^{1000}C_2 \cdot 5^2 + ...+ {}^{1000}C_{1000} \cdot 5^{1000} - 1$
Which is divisible by 5

10. Find the term independent of 'x' in the expansion of the expression $(1 + x + 2x^3)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$.

$(1 + x + 2x^3)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ के प्रसार में x' से स्वतंत्र पद ज्ञात कीजिए।

Ans. $\frac{17}{54}$

Sol. Co-efficient of x^0 in $(1 + x + 2x^3)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$

$= $ Co-efficient of x^0 in $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ + Co-efficient of x^{-1} in $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$

$+ 2$ Co-efficient of x^{-3} in $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$

$= {}^9C_{r_1}\left(\frac{3}{2}x^2\right)^{9-r_1}\left(-\frac{1}{3x}\right)^{r_1} + {}^9C_{r_2}\left(\frac{3}{2}x^2\right)^{9-r_2}\left(-\frac{1}{3x}\right)^{r_2} + 2\ {}^9C_{r_3}\left(\frac{3}{2}x^2\right)^{9-r_3}\left(-\frac{1}{3x}\right)^{r_3}$

$= {}^9C_6\left(\frac{3}{2}\right)^3\left(-\frac{1}{3}\right)^6 + 2\ {}^9C_7\left(\frac{3}{2}\right)^2\left(-\frac{1}{3}\right)^7 = \frac{17}{54}$

11. (i) Find the coefficient of x^5 in $(1 + 2x)^6(1 - x)^7$

(ii) Find the coefficient of x^4 in $(1 + 2x)^4 (2 - x)^5$

(i) $(1 + 2x)^6(1 - x)^7$ में x^5 का गुणांक ज्ञात कीजिए।

(ii) $(1 + 2x)^4 (2 - x)^5$ में x^4 का गुणांक ज्ञात कीजिए।

Ans. (i) 171 (ii) -438

Sol. (i) $(1 + 2x)^6 (1 - x)^7$

$= (1 + {}^6C_1 (2x) + {}^6C_2 (2x)^2 + {}^6C_3 (2x)^3 + {}^6C_4 (2x)^4 + {}^6C_5 (2x)^5 + (2x)^6) (1 - x)^7$

$= (1 + 12x + 60x^2 + 160x^3 + 240x^4 + 192x^5 + 64x^6) (1 - x)^7$

$= 1 \times$ coeff of $x^5 + 12 \times$ coeff. of $x^4 + 60 \times$ coeff of $x^3 + 160 \times$ coeff of $x^2 +$

$240 \times$ coeff of $x + 192 \times$ constant term.

$= 1 \times (-1)^5 \cdot {}^7C_5 + 12 \times {}^7C_4 - 60 \times {}^7C_3 + 160 \times {}^7C_2 - 240 \times {}^7C_1 + 192 \times 1$

$= -21 + 420 - 2100 + 3360 - 1680 + 192 = 171$

(ii) $(1 + 2x)^4 (2 - x)^5$

$[1 + {}^4C_1 (2x) + {}^4C_2 (2x)^2 + {}^4C_3 (2x)^3 + {}^4C_4(2x)^4] (2 - x)^5$

$= (1 + 8x + 24x^2 + 32x^3 + 16x^4)(2 - x)^5$

coefficient of $x^4 = 1 \times$ coefficient of $x^4 + 8 \times$ coefficient of $x^3 + 24 \times$ coefficient of $x^2 + 32 \times$ coefficient of $x + 16 \times$ constant term

$T_{r+1} = {}^5C_r \cdot {}^{25\text{-}r}(-x)_r = (-1)_r\, {}^5C_r\, 2^{5-r}\, x^r = 1 \times {}^5C_4 \times 2^1 - 8 \times {}^5C_3 \times 2^2 + 24 \times {}^5C_2 \times 2^3 - 32 \times {}^5C_1 \times 2^4 + 16 \times 2^5$

$= 10 - 320 + 1920 - 2560 + 512 = -438$

12. (i) Find numerically greatest term(s) in the expansion of $(3 - 5x)^{15}$ when $x = \frac{1}{5}$

(ii) Which term is the numerically greatest term in the expansion of $(2x + 5y)^{34}$, when $x = 3$ & $y = 2$?

(i) यदि $x = \frac{1}{5}$ तब $(3 - 5x)^{15}$ के प्रसार में महत्तम संख्यात्मक मान वाला (वाले) पद ज्ञात करो।

(ii) $(2x + 5y)^{34}$ के विस्तार में सख्यात्मक महत्तम पद होगा जब $x = 3$ तथा $y = 2$?

Ans. (i) $T4 = -455 \times 3^{12}$ and $T_5 = 455 \times 3^{12}$ (ii) 22

Sol. (i) For numerically greatest term in $(x + a)^n$ $r = \left[\frac{n+1}{1+\left|\frac{x}{a}\right|}\right]$

Since value of $\frac{n+1}{1+\left|\frac{x}{a}\right|}$ is itself an integer. There are two terms, whose numerical values are greatest

T_4 and T_5

$T4 = {}^{15}C_3\,(3)^{12}(-1)^3 = -455 \times 3^{12}$

$T_5 = {}^{15}C_4\,(3)^{11}(-1)^4 = 455 \times 3^{12}$

1. The coefficient of term that contains $x^{10}\,y^3\,z^2$ in the expansion of the binomial $(x^2-2y+z)^{10}$ is :
$(x^2-2y+z)^{10}$ के प्रसार में $x^{10}\,y^3\,z^2$ का गुणांक होगा:

(A) $-\dfrac{8\cdot(10)!}{2!\cdot3!\cdot5!}$ (B) $\dfrac{8\cdot(10)!}{2!\cdot3!\cdot5!}$ (C) $\dfrac{(10)!}{2!\cdot3!\cdot5!}$ (D) $-\dfrac{(10)!}{2!\cdot3!\cdot5!}$

Ans. A

Sol. ${}^0C_r(x^2)^{10-r}\,(2-2y)^r$

put $r=5$

${}^{10}C_5(x^2)^5\,(2-2y)^5\cdot{}^5C_p(2)^{r-p}\,(-2y)^p$

${}^{10}C_5\cdot x^{10}\,{}^5C_3(2)^2\cdot(2)^3\cdot y^3$

$-\dfrac{8\cdot(10)!}{2!\cdot3!\cdot5!}$

2. The coefficient of x^5 in the expansion of $(x^2-x-2)^5$ is :
$(x^2-x-2)^5$ के प्रसार में x^5 का गुणांक है :

(A) – 83 (B) – 82 (C) – 81 (D) 0

Ans. C

Sol. Coefficient of x^5 in the expansion of $(x^2-x-2)^5$

$$\underbrace{\left(x^2-x-2\right)^5=\sum_{k_1+k_2+k_3=5}\frac{\lfloor 5}{\lfloor k_1\lfloor k_2\lfloor k_3}x^{2k_1}\cdot(-x)^{k_2}\cdot(-2)^{k_3}}_{\text{Not helpful/long method}}$$

$(x^2-x-2)^5=[(x+1)(x-2)]^5$

$=(x+1)^5\,(x-2)^5=-(1+x)^5\,(2-x)^5$

$$=-\left[\sum_{r=0}^{5}{}^5C_r x^r\right]\left[\sum_{r=0}^{5}{}^5C_r(-x)^r\cdot2^{5-r}\right]$$

Coefficient of x^5 is

$$\left[{}^5C_0\cdot(-1)^5\,C_5+{}^5C_1\cdot{}^5C_4\cdot2+{}^5C_2\cdot(-1)^5\,C_3\cdot4+{}^5C_3\cdot{}^5C_2\cdot8+{}^5C_4\cdot{}^5C_1(-1)\cdot16+{}^5C_5\cdot{}^5C_0\cdot32\right]$$

$=-[-1+50+(-400)+800+(-400)+32]=-81$

3. In the expansion of $(1+3x+2x^2)^6$ the coefficient of x^{11} is :
$(1+3x+2x^2)^6$ के प्रसार में x^{11} का गुणांक है :

(A) 144 (B) 288 (C) 216 (D) 576

Ans. D

Sol. $\left(1+3x+2x^2\right)^6 = \left(x^2+\frac{3x}{2}+\frac{1}{2}\right)^6 \cdot 2^6$

$= (x+1)^6\left(x+\frac{1}{2}\right)^6 \cdot 2^6$

$= 2^6 \cdot \left[\sum_{r=0}^{6}(1+x)^6\right]\left[\left[\sum_{r=0}^{6}(1+2x)^6\right]\cdot\frac{1}{2^6}\right]$

$= \left[\sum_{r=0}^{6}(1+x)^6\right]\left[\sum_{r=0}^{6}(1+2x)^6\right]$

Coefficient of x^{11}

$= \left[{}^6C_5 \cdot {}^6C_6 (2)^6 + {}^6C_6 \cdot {}^6C_5 \cdot (2)^5\right]$

$= 6 \cdot 2^6 + 6 \cdot 2^5 = 6 \cdot 3 \cdot 2^5 = 576$

4. Let $\left(5+2\sqrt{6}\right)^n = p + f$ where $n \in N$ and $p \in N$ and $0 < f < 1$ then the value of, $f^2 - f + pf - p$ is

(A) a natural number (B) a negative integer
(C) a prime number (D) are irrational number

यदि $\left(5+2\sqrt{6}\right)^n = p + f$ जहाँ $p, n \in N$ तथा $0 < f < 1$ तो $f^2 - f + pf - p$ का मान होगा।

(A) एक प्राकृतिक संख्या (B) एक ऋणात्मक पूर्णांक
(C) एक अभाज्य संख्या (D) अपरिमेय संख्या

Ans. B

Sol. $f^2 - f + pf - p = -f(1-f) - p(1-f) = -[(p+f)(1-f)]$

where $p + f = (5+\sqrt{24})^n$ and $1 - f = f' = (5-\sqrt{24})^n$

hence $f^2 - f + pf - p = -1$

5. The greatest integer less than or equal to $\left(\sqrt{2}+1\right)^6$ is :

$\left(\sqrt{2}+1\right)^6$ से छोटा या बराबर महत्तम पूर्णांक है:

(A) 196 (B) 197 (C) 198 (D) 199

Ans. B

Sol. $\left(\sqrt{2}+1\right)^6 = \sum_{r=0}^{6} {}^6C_r \cdot \left(\sqrt{2}\right)^{6-r}$

$= {}^6C_0\left(\sqrt{2}\right)^6 + {}^6C_1\left(\sqrt{2}\right)^5 + {}^6C_2\left(\sqrt{2}\right)^4 + {}^6C_3\left(\sqrt{2}\right)^3 + {}^6C_4\left(\sqrt{2}\right)^2 + {}^6C_5\left(\sqrt{2}\right) + {}^6C_6$

$= \left(8+15\cdot4+30+1\right) + \sqrt{2}\left[{}^6C_1\left(\sqrt{2}\right)^4 + {}^6C_3\left(\sqrt{2}\right)^2 + {}^6C_5\right]$

$= 99 + \sqrt{2}\left[6\times4+20\cdot2+6\right]$

$= 99 + \sqrt{2}\left[70\right]$

$= 197 \cdot 98$

$\Rightarrow I = [197 \cdot 98] = 197$

6. If $\left(5+2\sqrt{6}\right)^n = I + f$, where $I \in N$, $n \in N$ and $0 \le f \le 1$, then I equals :

यदि $\left(5+2\sqrt{6}\right)^n = I + f$, जहाँ $I \in N$, $n \in N$ और $0 \le f \le 1$, तब I बराबर है :

(A) $\frac{1}{-f} - f$ (B) $\frac{1}{1+f} - f$ (C) $\frac{1}{1-f} - f$ (D) $\frac{1}{1-f} + f$

Ans. C

Sol. $\left(5+2\sqrt{6}\right)^n = I + f; I \in N \ \& \ 0 \le f \le 1$

$\left(5+2\sqrt{6}\right)^n = I + f = \sum_{r=0}^{n} {}^nC_r 5^{n-r}\left(2\sqrt{6}\right)^r$ ...(i)

$\left(5-2\sqrt{6}\right)^n = \left(5-\sqrt{24}\right)^n = 0 + f' = \sum_{r=0}^{n} {}^nC_r 5^{n-r}\left(-2\sqrt{6}\right)^r$ ...(ii)

Add $\Rightarrow I + f + f' = 2\left[{}^nC_0 5^n + {}^nC_2 5^{n-2}\left(2\sqrt{6}\right)^2 + ...\right]$

$\because 0 \le f \le 1, 0 \le f' \le 1$

$\Rightarrow 0 < f + f' < 2 \Rightarrow f + f' = 1$

$\Rightarrow I = 2\left[{}^nC_0 5^n + {}^nC_2 5^{n-2}\left(2\sqrt{6}\right)^2 + ...\right] - 1$

$\Rightarrow I + f + f' = 2I_1$

$\because f' = \left(5 - 2\sqrt{6}\right)^n = \dfrac{1}{\left(5 + 2\sqrt{6}\right)^n} = \dfrac{1}{I + f}$

$\Rightarrow f + f' = f' + \dfrac{1}{I + f}$

$\Rightarrow 1 = f' + \dfrac{1}{I + f}$

$\Rightarrow I + f = \dfrac{1}{1 - f'}$

$\Rightarrow I = \dfrac{1}{1 - f'} - f$

7. $(2n+1)(2n+3)(2n+5) \ldots\ldots (4n-1)$ is equal to :
$(2n+1)(2n+3)(2n+5) \ldots\ldots (4n-1)$ का मान होगा :

(A) $\dfrac{(4n)!}{2^n . (2n)! (2n)!}$ (B) $\dfrac{(4n)!\ n!}{2^n . (2n)! (2n)!}$ (C) $\dfrac{(4n)!\ n!}{(2n)! (2n)!}$ (D) $\dfrac{(4n)!\ n!}{2^n\ ! (2n)!}$

Ans. B

Sol. $E = (2n+1)(2n+3)(2n+5) \ldots\ldots (4n-1)$

Multiply numerator and denominator by $(2n+2)(2n+4) \ldots\ldots (4n)$ & also by $(2n)!$ and $n!$.

$$E = \frac{(2n)!\ (2n+1)(2n+2)(2n+3) \ldots\ldots (4n-1) \cdot 4n}{(2n)!\ (2n+2)(2n+4) \ldots\ldots (2n+2n)}$$

$$= \frac{(4n)! \quad \times \quad (n)!}{(2n)!\ 2^n \left[(n+1)(n+2) \ldots\ldots (2n)\right] n!} = \frac{(n!) \cdot (4n)!}{2^n \cdot \left((2n)!\right)^2} \quad \Rightarrow \text{B}$$

8. The sum of the co–efficients of all the even powers of x in the expansion of $(2x^2 - 3x + 1)^{11}$ is :
(A) $2 \cdot 6^{10}$ (B) $3 \cdot 6^{10}$ (C) 6^{11} (D) none
$(2x^2 - 3x + 1)^{11}$ के प्रसार में x की सम घात वाले गुणांकों का योग होगा :
(A) $2 \cdot 6^{10}$ (B) $3 \cdot 6^{10}$ (C) 6^{11} (D) इनमें से कोई नहीं

Ans. B

Sol. Let $(2x^2 - 3x + 1)^{11} = a_0 + a_1 x + a_2 x^2 + \ldots\ldots + a_{22} x^{22}$; to find $a_0 + a_2 + a_4 + \ldots\ldots + a_{22}$?

$S_E + S_0 = P(1) = 0$ – (1) where $P(x) = (2x^2 - 3x + 1)^{11}$

$S_E - S_0 = P(-1) = 6^{11}$ – (2)

$\Rightarrow \quad 2\,S_E = 6^{11} \quad \Rightarrow \quad S_E = 3 \cdot 6^{10}$

9. Set of value of r for which, ${}^{18}C_{r-2} + 2 \cdot {}^{18}C_{r-1} + {}^{18}C_r \geq {}^{20}C_{13}$ contains :
${}^{18}C_{r-2} + 2 \cdot {}^{18}C_{r-1} + {}^{18}C_r \geq {}^{20}C_{13}$ के संतुष्ट करने वाले r के कुल मानों की संख्या होगी।:
(A) 4 (B) 5 (C) 7 (D) 10

Ans. C

Sol. LHS ${}^{20}C_r \geq {}^{20}C_{13} \quad \Rightarrow \quad {}^{20}C_r \geq {}^{20}C_7 \quad \Rightarrow \quad r = 7, 8, 9, 10, 11, 12, 13 \quad \Rightarrow \quad C$

10. If $(1 + x - 3x^2)^{2145} = a_0 + a_1x + a_2x^2 + \ldots\ldots$ then $a_0 - a_1 + a_2 - a_3 + \ldots..$ ends with :
${}^{18}C_{r-2} + 2 \cdot {}^{18}C_{r-1} + {}^{18}C_r \geq {}^{20}C_{13}$ के संतुष्ट करने वाले r के कुल मानों की संख्या होगी:
(A) 1 (B) 3 (C) 7 (D) 9

Ans. B

Sol. Put $x = -1$; $(-3)^{2145} = a_0 - a_1 + a_2 - a_3 + \ldots..$; $-(3)^{2145} = -(3^4)^{536} \cdot 3 \quad \Rightarrow$ ends in 3

11. The number of values of 'r' satisfying the equation, ${}^{39}C_{3r-1} - {}^{39}C_{r^2} = {}^{39}C_{r^2-1} - {}^{39}C_{3r}$ is :

समीकरण ${}^{39}C_{3r-1} - {}^{39}C_{r^2} = {}^{39}C_{r^2-1} - {}^{39}C_{3r}$ को संतुष्ट करने वाले r के सारे मान की संख्या होगी :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $r = 3$ or 5 ; $r = 0$ is not possible

12. The largest real value for x such that $\sum_{k=0}^{4}\left(\frac{5^{4-k}}{(4-k)!}\right)\left(\frac{x^k}{k!}\right) = \frac{8}{3}$ is :

x का अधिकतम वास्तविक मान बताओ जबकि $\sum_{k=0}^{4}\left(\frac{5^{4-k}}{(4-k)!}\right)\left(\frac{x^k}{k!}\right) = \frac{8}{3}$ है :

(A) $2\sqrt{2}-5$ (B) $2\sqrt{2}+5$ (C) $-2\sqrt{2}-5$ (D) $-2\sqrt{2}+5$

Ans. A

Sol. $\sum_{k=0}^{4}\left(\frac{5^{4-k}}{(4-k)!}\right)\left(\frac{x^k}{k!}\right) = \sum_{k=0}^{4}\left(\frac{5^{4-k}}{(4-k)!}\right)\left(\frac{x^k}{k!}\right)\frac{4!}{4!} = \sum_{k=0}^{4}\frac{{}^4C_k \cdot 5^{4-k} \cdot x^k}{4!} = \frac{(5+x)^4}{4!}$

now $\frac{(5+x)^4}{4!} = \frac{8}{3} \Rightarrow \quad (5+x)^4 = 64$

$x + 5 = 2^{3/2} = 2\sqrt{2} \quad \Rightarrow \quad x = 2\sqrt{2} - 5$

13. If $\sum_{r=1}^{10} r(r-1)\ {}^{10}C_r = k \cdot 2^9$, then k is equal to :

यदि $\sum_{r=1}^{10} r(r-1)\ {}^{10}C_r = k \cdot 2^9$, तो k का मान होगा :

(A) 10 (B) 45 (C) 90 (D) 100

Ans. B

Sol. We have $\sum_{r=1}^{10} {}^{10}C_r \cdot r(r-1) = \sum_{r=1}^{10} \frac{10!}{r!(10-r)!} r(r-1) = 90\sum_{r=1}^{10} \frac{(8!)\, r(r-1)}{r(r-1)(r-2)!(10-r)!}$

$$= 90\sum_{r=1}^{10} \frac{8!}{(r-2)!(10-r)!} = 90\sum_{r=2}^{10} \frac{8!}{(r-2)!(10-r)!}$$

$$= 90[{}^8C_0 + {}^8C_1 + {}^8C_2 + \ldots\ldots\ldots + {}^8C_8]$$

$$= 90 \cdot 2^8 = 45 \cdot 2^9 \Rightarrow k = 45$$

14. The sum $\frac{\binom{11}{0}}{1} + \frac{\binom{11}{1}}{2} + \frac{\binom{11}{2}}{3} + \ldots\ldots\ldots\ldots\ldots + \frac{\binom{11}{11}}{12}$ equals $\left(\text{where } \binom{n}{r} \text{ denotes } {}^nC_r\right)$:

$\frac{\binom{11}{0}}{1} + \frac{\binom{11}{1}}{2} + \frac{\binom{11}{2}}{3} + \ldots\ldots\ldots\ldots\ldots + \frac{\binom{11}{11}}{12}$ का योग होगा। $\left(\text{जहाँ } \binom{n}{r} \text{ को निर्दिष्ट } {}^nC_r\right)$:

(A) $\frac{2^{11}}{12}$ (B) $\frac{2^{12}}{12}$ (C) $\frac{2^{11}-1}{12}$ (D) $\frac{2^{12}-1}{12}$

Ans. D

Sol. Consider the expansion of $(1+x)^{11} = {}^{11}C_0 + {}^{11}C_1 x + {}^{11}C_2 x^2 + \ldots\ldots + {}^{11}C_{11} x^{11}$

Integrating both sides between 0 and 1, we get

$$\frac{\binom{11}{0}}{1} + \frac{\binom{11}{1}}{2} + \frac{\binom{11}{2}}{3} + \ldots\ldots\ldots\ldots\ldots + \frac{\binom{11}{11}}{12} = \frac{2^{12}-1}{12}$$

15. For $2 \le r \le n$, $\binom{n}{r} + 2\binom{n}{r-1} + \binom{n}{r-2} =$

$2 \le r \le n$ के लिए $\binom{n}{r} + 2\binom{n}{r-1} + \binom{n}{r-2} =$

(A) $\binom{n+1}{r-1}$ (B) $2\binom{n+1}{r+1}$ (C) $2\binom{n+2}{r}$ (D) $\binom{n+2}{r}$

Ans. D

Sol. ${}^nC_r + 2\cdot{}^nC_{r-1} + {}^nC_{r-2}$

${}^nC_r + {}^nC_{r-1} + {}^nC_{r-1} + {}^nC_{r-2}$

${}^{n+1}C_r + {}^{n+1}C_{r-1}$

${}^{n+2}C_r$

16. Find the coefficient of t^8 in the expansion of $(1+2t^2-t^3)^9$

श्रेणी $(1+2t^2-t^3)^9$ के प्रसार में t^8 का गुणांक क्या होगा।

Ans. 2520

Sol. We have $\left((1+2t^2)-t^3\right)^9 = {}^9C_0(1+2t^2)^9 - {}^9C_1(1+2t^2)^8\cdot t^3 + {}^9C_2(1+2t^2)^7\cdot t^6$

$$- {}^9C_3(1+2t^2)^6\cdot t^9 + \ldots. - {}^9C_9(t^3)^9$$

$\therefore$ Coefficient of t^8 in the expansion of $(1+2t^2-t^3)^9$

$= {}^9C_0\left(\text{coefficient of } t^8 \text{ in } (1+2t^2)^9\right) - {}^9C_1\left(\text{coefficient of } t^5 \text{ in } (1+2t^2)^8\right)$

$$+ {}^9C_2\left(\text{coefficient of } t^3 \text{ in } (1+2t^2)^7\right)$$

$= {}^9C_0\cdot {}^9C_4\ 2^4 - 0 + {}^9C_2\cdot {}^7C_1\cdot 2 = \frac{9\cdot8\cdot7\cdot6}{4\cdot3\cdot2\cdot1}\cdot 16 + \frac{9\cdot8}{2\cdot1}\cdot 7\cdot 2 = 9\cdot8\cdot7\cdot5 = 2520$

Alternative : $T_{r+1} = {}^9C_r t^{2r}(2-t)^r$

Now coefficient of t^8 appears in $r=3, r=4$

$r=3 \Rightarrow {}^9C_3\ t^6(2-t)^3 = {}^9C_3 \times t^6 \times {}^3C_p\ 2^{3-p}(-t)^p$

Put $p=2 \Rightarrow {}^9C_3 \times {}^3C_2 \times 2^1 \times t^8 = 84 \times 6 \times t^8$

Simillarly when $r=4$

${}^9C_4\ t^8(2-t)^4 = {}^9C_4 \times t^8 \times {}^4C_p\ 2^{4-p}(-t)^p$

Put $p=0 \Rightarrow {}^9C_4 \times {}^4C_0 \times 2^4 \times t^8 = 126 \times 16 \times t^8$

Hence coefficient of t^8 in the expansion of $(1+2t^2-t^3)^9$ is $84\times6 + 126\times16$

$$= 504 + 2016 = 2520$$

1. Let $\binom{n}{k}$ represents the combination of 'n' things taken 'k' at a time, then the value of the sum

$$\binom{99}{97}+\binom{98}{96}+\binom{97}{95}+.....+\binom{3}{1}+\binom{2}{0} \text{ equals :}$$

यदि $\binom{n}{k} = {}^nC_R$ तो $\binom{99}{97}+\binom{98}{96}+\binom{97}{95}+.....+\binom{3}{1}+\binom{2}{0}$ का मान होगा :

(A) $\binom{99}{97}$ (B) $\binom{100}{98}$ (C) $\binom{99}{98}$ (D) $\binom{100}{97}$

Ans. D

Sol. We have $S = {}^2C_0 + {}^3C_1 + {}^4C_2 + {}^5C_3 + ... + {}^{99}C_{97}$ (writing in reverse order)

$S = \underbrace{{}^3C_0 + {}^3C_1} + {}^4C_2 + {}^5C_3 + ... + {}^{97}C_{95} + {}^{98}C_{96} + {}^{99}C_{97}$

$S = \underbrace{{}^4C_1 + {}^4C_2} + {}^5C_3 + ... + {}^{97}C_{95} + {}^{98}C_{96} + {}^{99}C_{97}$ (Using ${}^nn_r + {}^nn_{r-1} = {}^{n+1}n_r$)

$S = {}^5C_2 + {}^5C_3 + ... + {}^{97}C_{95} + {}^{98}C_{96} + {}^{99}C_{97}$

Similarly continuing in the same manner, we get

$S = {}^{99}C_{96} + {}^{99}C_{97} = {}^{100}C_{97} = \frac{100\cdot97\cdot98}{3\cdot2} = 161700$

2. If C_r stands for ${}^nC_r$, then the sum of first (n+1) terms of the series $a\,C_0 - (a+d)\,C_1 + (a+2d)\,C_2 - (a+3d)\,C_3 + ...$ is:

(A) $a/2^n$ (B) na (C) 0 (D) None of these

यदि C_r, ${}^nC_r$ को व्यक्त करता है तो श्रेणी $aC_0 - (a+d)C_1 + (a+2d)C_2 + (a+3d)\,C_3 + ..$ के पहले (n+1) पदों का योग है:

(A) $a/2^n$ (B) na (C) 0 (D) इनमें से कोई नहीं

Ans. C

Sol. $a\cdot{}^nC_0 - (a+d)C_1 + (a+2d)C_2 - (a+3d)^nC_3 + ...n \text{ terms}$

$$= \sum_{r=0}^{n}(-1)^r\,(a+rd)\cdot{}^nC_r$$

$$= a\sum_{r=0}^{n}(-1)^r \cdot {}^nC_r + d\sum_{r=0}^{n}(-1)^r \cdot r \cdot {}^nC_r$$

$$= a\left[C_0 - C_1 + C_2 - C_3 + ... + (-1)^n C_n\right] + d\left[-C_1 + 2C_2 - 3C_3 + ... + (-1)^n \cdot n \cdot C_n\right]$$

$$= 0$$

$$\left.\begin{aligned}&\because (1+x)^n = {}^nC_0 + {}^nC_1 x + ... + {}^nC_n x^n \\ &\Rightarrow (1-1)^n = C_0 - C_1 + C_2 - C_3 + ... + C_n(-1)^n = 0\end{aligned}\right\}$$

3. The sum of $C_0^2 - C_1^2 + C_2^2 - ....+ (-1)^n C_n^2$ where n is an even integer, is :
$C_0^2 - C_1^2 + C_2^2 - ....+ (-1)^n C_n^2$ का योग, n सम पूर्णांक के लिऐ है :

(A) ${}^{2n}C_n$ (B) $(-1)^n \ {}^{2n}C_n$

(C) ${}^{2n}C_{n-1}$ (D) $(-1)^{n/2} \cdot {}^nC_{n/2}$

Ans. D

Sol. $\sum_{r=0}^{n}(-1)^r C_r^2 = C_0^2 - C_1^2 + C_2^2 - C_3^2 + ... + (-1)^n C_n^2$

$$\because (1-x)^n = {}^nC_0 - {}^nC_1 x + ... + (-1)^n {}^nC_n x^n \qquad ...(i)$$

$$(x+1)^n = C_0 x^n + C_1 x^{n-1} + ... + C_n x^0 \qquad ...(ii)$$

(i) × (ii); we get

$$\left(1-x^2\right)^n = \left(\sum_{r=0}^{n}(-1)^{rn} C_r x^r\right)\left(\sum_{r=0}^{n} {}^nC_r x^{n-r}\right)$$

Comparing coefficient of x^n

$C_0^2 - C_1^2 + ... + (-1)^n C_n^2 =$ coefficient of x^n in $(1-x^2)^n$

Given n is even

$$C_0^2 - C_1^2 + ... + (-1)^n C_n^2 = (-1)^{\frac{n}{2}} \ {}^nC_{\frac{n}{2}}$$

4. If n = 11 then $(C_0^2 - C_1^2 + C_2^2 - C_3^2 + ... + (-1)^n (C_n^2)$ equals:

(A) $(-1)^5 . {}^{10}C_5$ (B) 0 (C) ${}^{10}C_6$ (D) None of these

यदि n = 11 हो तो $(C_0^2 - C_1^2 + C_2^2 - C_3^2 + ... + (-1)^n (C_n^2)$ बराबर होगा :

(A) $(-1)^5 . {}^{10}C_5$ (B) 0 (C) ${}^{10}C_6$ (D) इनमें से कोई नहीं

Ans. B

Sol. n = 11

$$C_0^2 - C_1^2 + C_2^2 - C_3^2 + ... + (-1)^n C_n^2 = 99$$

$$\because (1+x)^n = C_0 + C_1 x + C_2 x + ... + C_n x^n$$

$$(x-1)^n = C_0 x^n - C_1 x^{n-1} + C_2 x^{n-2} + ... + C_n (-1)^n \cdot x^0$$

$$\Rightarrow -(x^2-1)^n = \left(\sum_{r=0}^{11} C_r x^r\right)\left(\sum_{r=0}^{11} C_r x^{11-r}\right)$$

Coefficient of x^{11} in the expansion of $(x^2-1)^{11}$ is 0.

5. The coefficient of x^5 in the expansion of $(1 + x)^{21} + (1 + x)^{22} + ...+ (1 + x)^{30}$ is :

$(1+x)^{21} + (1+x)^{22} + ...+ (1+x)^{30}$ के प्रसार में x^5 का गुणांक है :

(A) ${}^{51}C_5$ (B) ${}^{9}C_5$ (C) ${}^{31}C_6 - {}^{21}C_6$ (D) ${}^{30}C_5 + {}^{20}C_5$

Ans. C

Sol. Coefficient of x^5 in the expansion of $(1+x)^{21} + (1+x)^{22} + ... + (1+x)^{30}$

$= (1 + x)^{21} [1 + (1 + x) + ... + (1 + x)^9]$

$$= (1+x)^{21}\left[\frac{(1+x)^{10} - 1}{x}\right]$$

$$= \frac{(1+x)^{31}}{x} - \frac{(1+x)^{21}}{x}$$

Coefficient of x^5 is ${}^{31}C_6 - {}^{21}C_6$

6. If x + y = 1, then $\sum_{r=0}^{n} r^2 {}^{n}C_r x^r y^{n-r}$ equals :

(A) nxy (B) nx (x+ yn) (C) nx(nx + y) (D) None of these

यदि $x + y = 1$, तब $\sum_{r=0}^{n} r^2 {}^nC_r x^r y^{n-r}$ बराबर है :

(A) nxy (B) nx (x+ yn) (C) nx(nx + y) (D) इनमें से कोई नहीं

Ans. C

Sol. $\sum_{r=0}^{n} r^2 {}^nC_r \; p^r q^{n-r} = n \cdot \sum_{r=0}^{n} r \cdot {}^{n-1}C_{r-1} p^r q^{n-r}$

$$= n. \left[\sum_{r=1}^{n} (r-1) \cdot {}^{n-1}C_{r-1} p^r q^{n-r} + \sum_{r=1}^{n} {}^{n-1}C_{r-1} \cdot p^r q^{n-r} \right]$$

$$= n \left[(n-1)p^2 \sum_{r=2}^{n} {}^{n-2}C_{r-2} p^{r-2} q^{n-r} + p \sum_{r=1}^{n} {}^{n-1}C_{r-1} \cdot p^{r-1} q^{n-r} \right]$$

$= n[(n-1)p^2(p+q)^{n-2} + p(p+q)^{n-1}]$

$= n[np^2 - p^2 + p] = n^2p^2 - np^2 + pn = n^2p^2 + npq$

7. The value of ${}^{95}C_4 + \sum_{j=1}^{5} {}^{100-j}C_3$ is :

${}^{95}C_4 + \sum_{j=1}^{5} {}^{100-j}C_3$ का मान है :

(A) ${}^{95}C_5$ (B) ${}^{100}C_4$ (C) ${}^{99}C_4$ (D) ${}^{100}C_5$

Ans. B

Sol. ${}^{95}C_4 + \sum_{j=1}^{5} {}^{100-j}C_3$

$= {}^{95}C_4 + {}^{95}C_3 + {}^{96}C_3 + {}^{97}C_3 + {}^{98}C_3 + {}^{99}C_3$

$= {}^{96}C_4 + {}^{96}C_3 + {}^{97}C_3 + {}^{98}C_3 + {}^{99}C_3$

$\left\{ \because {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r \right\}$

$\vdots$

$= {}^{100}C_4$

8. If $\left({}^{15}C_r + {}^{15}C_{r-1}\right) \left({}^{15}C_{15-r} + {}^{15}C_{16-r}\right) = \left({}^{16}C_{13}\right)^2$, then the value of r is :

(A) r = 3 (B) r = 2 (C) r = 4 (D) None of these

यदि $\left({}^{15}C_r + {}^{15}C_{r-1}\right) \left({}^{15}C_{15-r} + {}^{15}C_{16-r}\right) = \left({}^{16}C_{13}\right)^2$, तब r का मान है -

(A) r = 3 (B) r = 2 (C) r = 4 (D) इनमें से कोई नहीं

Ans. A

Sol. $\left({}^{15}C_r + {}^{15}C_{r-1}\right)\left({}^{15}C_{15-r} + {}^{15}C_{16-r}\right) = \left({}^{16}C_{13}\right)^2$

$\Rightarrow \left({}^{16}C_r\right)\left({}^{15}C_{16-r} + {}^{15}C_{15-r}\right) = \left({}^{16}C_{13}\right)^2$

$\left\{\because {}^{n}C_r + {}^{n}C_{r-1} = {}^{n+1}C_r\right\}$

$\Rightarrow \left({}^{16}C_r\right)\left[{}^{16}C_{16-r}\right] = \left({}^{16}C_{13}\right)^2$

$\Rightarrow \left[{}^{16}C_r\right]^2 = \left[{}^{16}C_{13}\right]^2 \Rightarrow \left({}^{16}C_r\right)^2 = \left({}^{16}C_3\right)^2$

$\Rightarrow r = 13 \left\{\because {}^{16}C_{13} = {}^{16}C_{13}\right\}$

9. In the expansion of $(1 + x + x^2 + ... + x^{27})(1 + x + x^2 + ... + x^{14})^2$, the coefficient of x^{28} is:
$(1 + x + x^2 + ... + x^{27})(1 + x + x^2 + ... + x^{14})^2$ के प्रसार में x^{28} का गुणांक होगा
(A) 195 (B) 224 (C) 378 (D) 405

Ans. B

Sol. Concept: Coefficient of x^r in $(1-x)^{-n}$, $n \in N$ is ${}^{n+r-1}C_r$.

now given product is $= \frac{1-x^{28}}{1-x} \cdot \left(\frac{1-x^{15}}{1-x}\right)^2$

$$= \frac{(1-x^{28})(1-x^{15})^2}{(1-x)^3} = \frac{(1-x^{28})(1-2x^{15})}{(1-x)^3} = (1 - 2x^{15} - x^{28})(1-x)^{-3}$$

Hence coefficient of x^{28} $(1 - 2x^5 - x^{28})(1-x)^{-3} - 2 \cdot$ coefficient of x^{13} in $(1-x)^{-3} - 1$
$= {}^{30}C_2 - 2 \cdot {}^{15}C_2 - 1 = 435 - 210 - 1 = 224$

Paragraph for question nos. 10 to 12

If $n \in N$ and if $(1 + 4x + 4x^2)^n = \sum_{r=0}^{2n} a_r x^r$, where $a_0, a_1, a_2, ..., a_{2n}$ are real numbers.

यदि $(1 + 4x + 4x^2)^n = \sum_{r=0}^{2n} a_r x^r$ जहाँ $n \in N$ एवं $a_0, a_1, a_2, ..., a_{2n}$ वास्तविक संख्या हो, तो

10. The value of $2\sum_{r=0}^{n} a_{2r}$, is :

$2\sum_{r=0}^{n} a_{2r}$ का मान होगा :

(A) $9^n - 1$ (B) $9^n + 1$ (C) $9^n - 2$ (D) $9^n + 2$

Ans. B

Sol. $(1 + 2x)^{2n} = a_0 + a_1x + a_2x^2 + ... + a_{2n-1}x^{2n-1} + a_{2n}x^{2n}$

$x = 1$ $3^{2n} = a_0 + a_1 + a_2 + ... + a_{2n-1} + a_{2n}$ ....(1)

$x = -1$ $1 = a_0 - a_1 + a_2 - ..... + a_{2n}$ ....(2)

(1) + (2) $3^{2n} + 1 = 2[a_0 + a_2 + a_4 + ... + a_{2n}]$

$$9^n + 1 = 2\sum_{r=0}^{n} a_{2r}$$

11. The value of $2\sum_{r=1}^{n} a_{2r-1}$, is :

$2\sum_{r=1}^{n} a_{2r-1}$ का मान होगा:

(A) $9^n - 1$ (B) $9^n + 1$ (C) $9^n - 2$ (D) $9^n + 2$

Ans. A

Sol. $(1 + 2x)^{2n} = a_0 + a_1x + a_2x^2 + ... + a_{2n-1}x^{2n-1} + a_{2n}x^{2n}$

$x = 1$ $3^{2n} = a_0 + a_1 + a_2 + ... + a_{2n-1} + a_{2n}$ ....(1)

$x = -1$ $1 = a_0 - a_1 + a_2 - ..... + a_{2n}$ ....(2)

(1) – (2) $3^{2n} - 1 = 2[a_1 + a_3 + a_5 + ... + a_{2n-1}]$

$$9^n - 1 = 2\sum_{r=1}^{n} a_{2r-1}$$

12. The value of a_{2n-1} is :

a_{2n-1} का मान होगा:

(A) 2^{2n} (B) $n \cdot 2^{2n}$ (C) $(n - 1)2^{2n}$ (D) $(n + 1)2^{2n}$

Ans. B

Sol. $(1 + 2x)^{2n} = a_0 + a_1x + a_2x^2 + ... + a_{2n-1}x^{2n-1} + a_{2n}x^{2n}$

$x = 1$ $3^{2n} = a_0 + a_1 + a_2 + ... + a_{2n-1} + a_{2n}$ ....(1)

$x = -1$ $1 = a_0 - a_1 + a_2 - ..... + a_{2n}$ ....(2)

a_{2n-1} = coefficient x^{2n-1} in $(1 + 2x)^{2n}$

$T_{r+1} = {}^{2n}C_r (2x)^r = {}^{2n}C_r \cdot 2^r x^2$

$r = 2n - 1$, ${}^{2n}C_{2n-1}\, 2^{2n-1}$

$2n \cdot 2^{2n-1}$

$n \cdot 2^{2n}$

13. **Statement-1:** The expression $n!(100-n)!$ is maximum when $n = 50$.

Statement-2: ${}^{2n}C_r$ is maximum when $r = n$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false. (D) Statement-1 is false, statement-2 is true

कथन-1: व्यंजक $n!(100-n)!$ का अधिकतम मान $n = 50$ पर होगा

कथन-2: ${}^{2n}C_r$ का अधिकतम मान $r = n$ पर होगा

(A) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण है।

(B) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण **नहीं** है।

(C) कथन–1 सत्य है, कथन–2 असत्य है।

(D) कथन–1 असत्य है, कथन– 2 सत्य है।

Ans. D

Sol. We have $n!\,(100-n)! = \dfrac{100!}{{}^{100}C_n}$

Now ${}^{2n}C_r$ is maximum when $r = n$

$\therefore \quad n!\,(100-n)! = \dfrac{100!}{{}^{100}C_n}$

Hence $n!(100-n)!$ will be minimum when $n = 50$

14. If C_r denotes the combinatorial coefficient in the expansion of $(1+x)^n$ then $(n \in N)$

Statement-1: $(n+1)\left(C_0^2 + C_1^2 + C_2^2 + \ldots\ldots + C_n^2\right) > \left(C_0 + C_1 + C_2 + \ldots\ldots + C_n\right)^2 \;\forall\; n \geq 2,\ n \in N$

because

Statement-2: Root mean square of *n* distinct real numbers is always greater than their arithmetic mean, $n \geq 2$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true

${}^nC_r$ द्विपद $(1+x)^n$ के प्रसार में सारे द्विपद गुणांक है जहाँ $(n \in N)$

कथन-1: $(n+1)\left(C_0^2 + C_1^2 + C_2^2 + \ldots\ldots + C_n^2\right) > \left(C_0 + C_1 + C_2 + \ldots\ldots + C_n\right)^2 \;\forall\; n \geq 2,\ n \in N$

क्योंकि

कथन-2: *n* भिन्न–भिन्न धनात्मक संख्याओं का वर्ग माध्य मूल हमेशा उनके समान्तर माध्य से बड़ा होगा (जहाँ $n \geq 2$)

(A) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण है।

(B) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण **नहीं** है।

(C) कथन–1 सत्य है, कथन–2 असत्य है।

(D) कथन–1 असत्य है, कथन– 2 सत्य है।

Ans. A

Sol. Since root mean square of n distinct real number is always greater than their arithmetic mean. ($n \geq 2$)

$$\sqrt{\frac{C_0^2 + C_1^2 + ... + C_n^2}{n+1}} > \frac{C_0 + C_1 + ... + C_n}{n+1}$$

$$\Rightarrow \frac{C_0^2 + C_1^2 + ... + C_n^2}{n+1} > \frac{(C_0 + C_1 + ... + C_n)^2}{(n+1)^2}$$

$$\Rightarrow (n+1)\left[C_0^2 + C_1^2 + ... + C_n^2\right] > (C_0 + C_1 + ... + C_n)^2$$

(A) is right.

15. Statement-1: The sum of the series
$^nc_0 \cdot {}^mc_r + {}^nc_1 \cdot {}^mc_{r-1} + {}^nc_2 \cdot {}^mc_{r-2} + ... + {}^nc_r \cdot {}^mc_0$ is equal to $^{n+m}c_r$.
where ^nC_r's and ^mC_r's denotes the combinatorial coefficients in the expansion of $(1 + x)^n$ and $(1 + x)^m$ respectively.
because
Statement-2: Number of ways in which *r* children can be selected out of (n + m) children consisting of *n* boys and *m* girls if each selection may consists of any number of boys and girls is equal to $^{n+m}C_r$.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false
(D) Statement-1 is false, statement-2 is true
कथन-1: श्रेणी $^nc_0 \cdot {}^mc_r + {}^nc_1 \cdot {}^mc_{r-1} + {}^nc_2 \cdot {}^mc_{r-2} + ... + {}^nc_r \cdot {}^mc_0$ का योग $^{n+m}c_r$ होगा।
जहाँ ^nC_r's तथा ^mC_r's क्रमशः श्रेणी $(1 + x)^n$ तथा $(1 + x)^m$ के प्रसार में गुणांक है
क्योंकि
कथन-2: n लड़के तथा m लड़कियों के समूह (n + m) में से जिसे *r* बच्चे को चुनने का तरीका $^{n+m}C_r$ होगा।
(A) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण है।
(B) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण **नहीं** है।
(C) कथन–1 सत्य है, कथन–2 असत्य है।
(D) कथन–1 असत्य है, कथन– 2 सत्य है।

Ans. A

Sol. Note : It is Vandermonde's Identity.

Let there be 'n' boys and 'm' girls. We want to choose a team of 'r' persons, the boy-girl proportion allowed to be all possible. The 'r' persons can be chosen from n + m persons in $^{n+m}c_r$ ways.
Also, number of ways ;
when no boy and 'r' girls = $^nc_0 \cdot {}^mc_r$
when 1 boy and r – 1 girls = $^nc_1 \cdot {}^mc_{r-1}$
Similarly 'r' boys and no girls = $^nc_r \cdot {}^mc_0$
Hence, total = $^nc_0 \cdot {}^mc_r + {}^nc_1 \cdot {}^mc_{r-1} + ... + {}^nc_r \cdot {}^mc_0 = {}^{m+n}c_r$
Alternatively : compare the co-efficient of x^r in $(1 + x)^{m+n}$ and $(1 + x)^m \cdot (1 + x)^n$

16. If the sum $S = {}^{20}C_2 + 2 \cdot {}^{20}C_3 + 3 \cdot {}^{20}C_4 + \ldots\ldots\ldots\ldots + 19 \cdot {}^{20}C_{20}$ can be expressed as $1 + p \cdot 2^q$, then find the minimum value of $(p + q)$ where $p, q \in N$.

यदि $S = {}^{20}C_2 + 2 \cdot {}^{20}C_3 + 3 \cdot {}^{20}C_4 + \ldots\ldots\ldots\ldots + 19 \cdot {}^{20}C_{20}$ का मान $1 + p \cdot 2^q$, है, तो $(p + q)$ का न्यूनतम मान होगा जहां $p, q \in N$ है।

Ans. 29

Sol. Let $S = {}^{20}C_2 + 2 \cdot {}^{20}C_3 + 3 \cdot {}^{20}C_4 + \ldots\ldots\ldots\ldots + 19 \cdot {}^{20}C_{20}$ ....(1)

add ${}^{20}C_0 + {}^{20}C_1 + \ldots\ldots\ldots\ldots + {}^{20}C_{20} (= 2^{20})$ on both sides of equation (1)

$$S + 2^{20} = {}^{20}C_2 + 2 \cdot {}^{20}C_3 + 3 \cdot {}^{20}C_4 + \ldots\ldots\ldots\ldots + 19 \cdot {}^{20}C_{20}$$
$$+ {}^{20}C_0 + {}^{20}C_1 + {}^{20}C_2 + {}^{20}C_3 + {}^{20}C_4 + \ldots\ldots\ldots\ldots + {}^{20}C_{20}$$

$S + 2^{20} = 1 + {}^{20}C_1 + 2 \cdot {}^{20}C_2 + 3 \cdot {}^{20}C_3 + 4 \cdot {}^{20}C_4 + \ldots\ldots\ldots\ldots + 20 \cdot {}^{20}C_{20}$ ....(2)

now

$(1 + x)^n = C_0 + C_1 x + C_2 x^2 + \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots + C_n x^n$

$n(1 + x)^{n-1} = C_1 + 2 \cdot C_2 x + 3 \cdot C_3 x^2 + \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots + n \cdot C_n x^{n-1}$

put $x = 1$ and $n = 20$

$20 \cdot 2^{19} = {}^{20}C_1 + 2\,{}^{20}C_2 + 3\,{}^{20}C_3 + \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots + 20\,{}^{20}C_{20}$

hence $S + 2^{20} = 1 + 20 \cdot 2^{19}$

$S = 1 + 20 \cdot 2^{19} - 2^{20} = 1 + 10 \cdot 2^{20} - 2^{20} = 1 + 9 \cdot 2^{20s}$

17. (i) Find the coefficient of x^{12} in $\dfrac{4 + 2x - x^2}{(1+x)^3}$

(ii) Find the coefficient of x^{100} in $\dfrac{3 - 5x}{(1-x)^2}$

(i) $\dfrac{4 + 2x - x^2}{(1+x)^3}$ में x^{12} का गुणांक ज्ञात कीजिए।

(ii) $\dfrac{3 - 5x}{(1-x)^2}$ में x^{100} का गुणांक ज्ञात कीजिए।

Ans. (i) 142

(ii) – 197

Sol. (i) $(4 + 2x - x^2)(1 + x)^{-3}$

$= 4 \times$ coeff of $x^{12} + 2 \times$ coeff of $x^{11} - 1 \times$ coeff of x^{10}

In $(1 + x)^{-3}$

$T_{r+1} = (-1)^r \ {}^{3+r}C_1 C_r x^r$

$= (-1)^r \ {}^{r+2}C_r \, xr$

$= 4 \times {}^{14}C_{12} - 2 \times {}^{13}C_{11} - {}^{12}C_{10}$

$= 4 \times \frac{14 \times 13}{2} - 2x \frac{13 \times 12}{2} - \frac{12 \times 11}{2}$

$= 364 - 156 - 66 = 142$

(ii) $(3 - 5x)(1 - x)^{-2}$

$= 3 \times \text{coeff of } x^{100} - 5 \times \text{coeff of } x^{99}$

$\text{In } (1 - x)^{-2}$

$T_{r+1} = {}^{2+r-1}C_r \, x^r = {}^{r+1}C_r \, x^r = 3 \times {}^{101}C_{100} - 5 \, x \, {}^{100}C_{99} = 3 \times 101 - 5 \times 100 = -197$